

GRAFT: Symmetry-Aware Atom Mapping through Continuous Fragment Growth

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Atom-to-atom mapping for reaction analysis requires retaining alternative correspondences without enumerating redundant atom permutations. We introduce GRAFT, a graph-based method that grows matched fragments on weighted reactant and product graphs, represents compatible placements through correlated symmetry actions, and branches both on saturated placements and on competition between fragments. Completed searches are deduplicated by unordered matched fragment pairs while preserving their internal mapping alternatives. A symbolic decoder extracts distinct signed bond-event patterns directly from these families, separating the tolerance used for matching from the thresholds used to interpret bond changes. On all 1,851 Golden benchmark records, bidirectional search with edge sweeps recovers an annotated correspondence among its retained alternatives in 1,834 cases (99.08%) with one seed ordering per cut (default) and 1,840 (99.41%) with ten. All GRAFT reference-recovery verdicts are resolved. The strongest evaluated prior-method alternative-coverage baseline, the bidirectional SLAP sweep, recovers 97.03%. On the matched 1,821-reaction timing cohort, default GRAFT and SLAP sweep average 2.52 and 4.82 CPU seconds per reaction, respectively. On 140 coordinate/Wiberg-bond-order reaction pairs, GRAFT reaches the same minimum event count as the SLAP sweep in 136 reactions and fewer events in four. Where their minima agree, it retains 162 of 164 comparator minimum-event patterns. This bond-event comparison uses unverified mappings and does not measure mapping accuracy. GRAFT provides an inspectable representation of mapping and bond-event alternatives for subsequent chemical assessment.

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1 Introduction

Atom-to-atom mapping assigns corresponding atoms across a reaction. It underlies reaction-center identification, template extraction, and the preparation of endpoint structures for mechanistic calculations. A

reaction can admit several plausible correspondences, while many different atom-index assignments are chemically equivalent. An algorithm intended to retain alternatives must therefore resolve two questions: which structural choices deserve separate candidates, and which permutations can be represented together?

Existing approaches use learned correspondence signals or graph-based optimization. RXNMapper derives mappings from attention in a reaction language model (11); LocalMapper learns from chemist-labeled reactions (2); and SLAPMapper combines iterative label refinement with sequential linear assignment (7). The Golden benchmark provides a common set of annotated reactions for assessing mapping recovery (8). For applications requiring alternatives, recovering the annotation somewhere in a returned family is a different objective from producing one correct top-ranked mapping.

Weighted endpoint graphs introduce a further distinction. A permissive matching tolerance can preserve a fragment despite changes in its bond orders, but the corresponding atom shuffles need not preserve the bond events assigned at a finer threshold. Replacing a compressed family by one representative can therefore hide distinct transformations. Likewise, greedily growing one fragment can prevent a neighboring fragment from reaching an alternative assignment.

GRAFT addresses these issues by branching over structural fragment choices while keeping symmetry-related atom bijections compressed (Figure 1). Continuous growth, alternative placements, and fragment priorities determine the structural choices. Its final representation groups unordered matched fragment pairs and retains the allowed mapping alternatives within each group. A constraint-based decoder projects these families onto distinct bond-event patterns. The method searches for structurally compatible alternatives without requiring the reference transformation to minimize the number of bond changes.

2 Method

2.1 Weighted matching and continuous growth

Let V_R, V_P be the reactant and product atom sets, Z_R, Z_P their element labels, and W_R, W_P their symmetric bond-weight matrices. A partial mapping $m : D \subseteq V_R \rightarrow V_P$ is injective and element compatible. A mapping assigning every source atom is bijective when the endpoint element counts agree. Explicit hydrogen atoms participate in the search. Molecular coordinates can accompany the graphs but do not determine the growth decisions.

The active source graph contains edges with weight at least b . Starting from a seed atom, GRAFT grows a connected source fragment F while carrying multiple compatible product placements. A frontier queue prioritizes stronger source edges. To add an unassigned source atom u with product image v , a placement must satisfy

$$Z_R(u) = Z_P(v), \quad v \notin m(D), \quad W_P(v, m(r)) \geq b, \quad |W_R(u, r) - W_P(v, m(r))| \leq \tau_{\text{iso}} \quad (1)$$

for every active, nonrelaxed source edge (u, r) into F . Injectivity is enforced against all current assignments in D , including established fragments. All relevant fragment bonds are checked, rather than only the frontier edge that proposed the extension. Source nonedges do not impose negative matching constraints, so additional product bonds remain possible.

An accepted extension enlarges the fragment and filters its compatible placements. Growth continues even if only one placement remains. An unsupported edge is deferred, and other frontier edges can still be explored. When growth reaches an established fragment, ordinary continuation respects its existing assignments. A fragment saturates when no further admissible extension remains; structurally distinct surviving placements then generate separate continuations for the remaining atoms.

This is a greedy search along each trajectory: an accepted extension can remove placements that would support a different fragment boundary. GRAFT broadens coverage through seed orderings, single-edge constraint relaxations, and local changes in fragment priority. A sweep combines the uncut source graph with searches in which one source edge with weight at least b is omitted, including hydrogen-containing edges in the evaluated configurations. Bond-event evaluation always uses the original endpoint matrices.

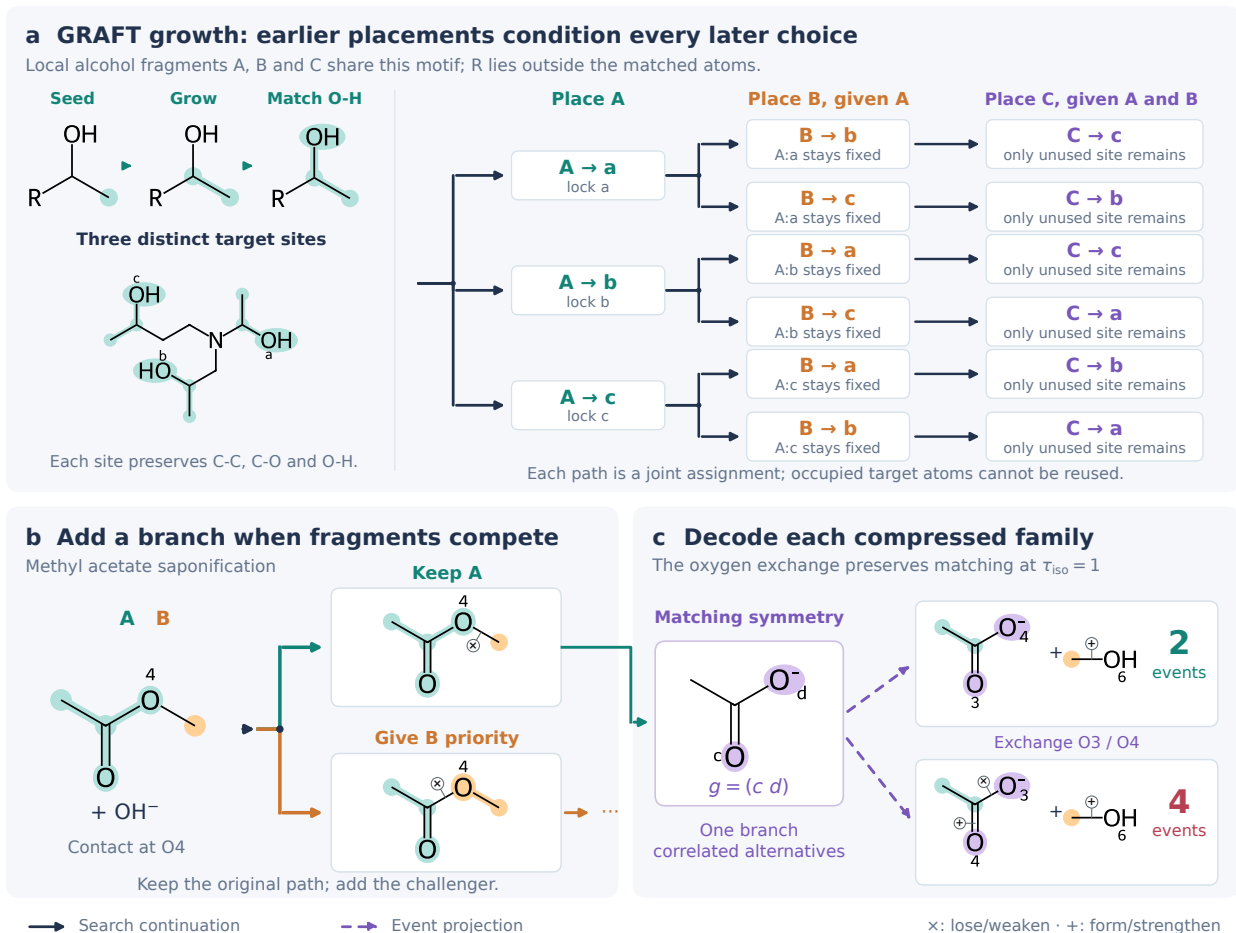


Figure 1 GRAFT: conditional fragment branching and compressed symmetry. (a) A local alcohol motif grows while retaining its O-H bond. Three sites in the depicted target have distinct molecular environments. For three source fragments sharing this motif, placing A fixes one site; B then branches over the two unused sites, and C takes the remaining site. The tree illustrates local conditioning and injectivity; R denotes atoms outside the matched subgraph, and this is not a complete reaction-search trajectory. All attached hydrogens in the illustrated local matches are checked. (b) At a fragment contact in methyl acetate saponification, the original path retains O4 in A. An added path gives B priority, rematches the contact, and regrows displaced assignments. (c) Exchanging the two acetate oxygen images preserves the fragment’s element and bond-weight matching tests at $\tau_{iso} = 1$, but changes two heavy-atom events into four at threshold 0.5. Neither acetate oxygen carries H. Solid black lines retain the endpoint bonds. Separate × and + badges mark cleavage/weakening and formation/strengthening, respectively; short leaders identify the affected bonds. Pastel colors track fragment membership or symmetry sites. Dashed arrows are event projections within a family, not additional search branches. Saved constraints govern allowed symmetry actions. These formal-bond-order event signatures illustrate GRAFT decoding, not distinct mechanisms or resonance-normalized transformations.

2.2 Automorphisms and correlated mapping families

The distinction between structural branches and explicit bijections is central to the design. Even one fragment family can contain combinatorially many assignments: k independently exchangeable methyl-hydrogen triplets admit $(3!)^k$ bijections. Expanding those automorphisms into separate mappings would multiply the work without adding fragment choices. GRAFT instead stores the permitted permutation actions and their constraints, and decodes event alternatives directly from this representation. Branch caps bound retained alternatives within each growth call; they do not guarantee a small global search. This avoids explicitly listing symmetry-related bijections within each retained family.

Compatible placements are stored using a representative assignment, unresolved assignment blocks, and permutation generators. During growth, graph canonicalization accounts for the current fragment, locked

assignments, and boundary information. Each saved transition records its action generators together with conditioning and fragment-preservation constraints; their conjunction defines the allowed mapping relation. Canonical labeling and automorphism computation use nauty (9).

Matching tolerance and automorphism equivalence have different roles. The relation $|x - y| \leq \tau_{\text{iso}}$ is generally not transitive, so it does not by itself define a permutation group. To describe an intrinsic symmetry shared by a final fragment pair, we instead color product bonds by their responses to the fragment’s matching tests. For a product weight w and the set $\mathcal{W}$ of active reactant weights for the same element pair, the label records

$$\chi_{\mathcal{W}}(w) = (\mathbf{1}[w \geq b], (\mathbf{1}[w \geq b \wedge |w - v| \leq \tau_{\text{iso}}])_{v \in \mathcal{W}}). \quad (2)$$

An exact automorphism of this colored graph preserves every encoded matching test. This definition depends on the final fragment pair, not its growth order.

The saved mapping relation also contains the constraints imposed by its search decisions. We represent a flat family $\mathcal{F}_j$ by a seed mapping m_j , a sequence of allowed action groups $G_{j,1}, \dots, G_{j,k}$, and final preservation constraints C_j :

$$\mathcal{F}_j = \{a_k \circ \dots \circ a_1 \circ m_j : a_i \in G_{j,i}, C_j(a_k \circ \dots \circ a_1 \circ m_j)\}. \quad (3)$$

These actions encode correlated moves; atoms in a common orbit are not independently permutable. Disjoint actions commute and can be placed in a canonical order. The order of overlapping actions is retained. A sequence of groups is not generally equivalent to the group generated by their union.

The intrinsic automorphism of a fragment pair is shared metadata, not permission to enlarge its saved family. Multiple admissible seed mappings can occupy different intrinsic symmetry orbits, and different search constraints can restrict the allowed actions. GRAFT therefore retains these alternatives explicitly inside the compressed representation.

2.3 Branching through fragment competition

Saturated-placement branching explores different images of a fragment. Fragment competition additionally explores a different division of the source atoms. Suppose established fragment A borders fragment B . In a challenger branch, GRAFT rematches B together with a contacting atom of A , allowing the challenger to occupy a different compatible product region. Assignments conflicting with the new placement are released. Unaffected assignments remain anchored, and the fragment-growth procedure fills the resulting gaps. Small attached components can also be released when their assignments depend on the displaced region.

The repaired continuation begins from the revised assignments and constraints; it does not inherit displaced assignments as immutable conditions. Before admission, its complete representatives are checked for element compatibility, injectivity, and preservation of their required fragment bonds. Accepted continuations join the original output, so competition supplements the available alternatives. A finite queue, a bound on successive priority changes, and a completion budget control its cost. Repeated proposals and equivalent completion requests are cached.

2.4 Unordered final branches

During search, history and boundary conditions matter because they determine valid continuation. At completion, the branch identity is instead the unordered collection of matched atom sets,

$$\mathcal{B} = \{(R_1, P_1), \dots, (R_q, P_q)\}, \quad P_i = m(R_i), \quad (4)$$

under a fixed matching policy. Sorting atoms within each set and then sorting the fragment pairs removes dependence on growth order and fragment labels. Growing A before B or B before A yields one branch when the final paired sets agree.

Each branch stores the union of its flat families. Identical normalized family records are shared, while distinct constraints or action programs remain available. This transformation preserves the represented mapping union: it changes its grouping, not its admissible assignments. Provenance links retain the discovering paths, but the original search hierarchy is unnecessary for decoding. This conservative deduplication does not assert that the remaining families are mutually disjoint.

2.5 Decoding distinct bond-event patterns

For each complete mapping, define the signed change for an unordered source pair $\{u, v\}$ by

$$\Delta_{uv}(m) = W_P(m(u), m(v)) - W_R(u, v), \quad s_{uv}(m) = \begin{cases} +1, & \Delta_{uv}(m) \geq \delta_{uv}, \\ -1, & \Delta_{uv}(m) \leq -\delta_{uv}, \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

Positive events denote bond formation or strengthening; negative events denote cleavage or weakening. The event count is $E(m) = \sum_{u < v} \mathbf{1}[s_{uv}(m) \neq 0]$. The evaluation includes hydrogen atoms and all atom pairs. For the WBO collection, $\delta_{uv} = 0.3$ for pairs involving a metal and 0.5 otherwise. These thresholds are independent of τ_{iso} .

The decoder returns one mapping witness per distinct signed event class, identifying patterns related by exact source permutations that preserve all thresholded bond-change tests. A permutation preserving fragment matching may therefore produce a distinct event class. First, symmetry-invariance certificates identify families whose mappings all produce equivalent events, and reachable-image bounds exclude families outside a requested event window. Before constraint encoding, adjacent nested action groups are absorbed: if $H \leq G$, then $HG = GH = G$. Exact generator-membership tests remove these redundant choices without changing the final preservation constraints. Remaining families are encoded symbolically in Z3 (5). Each satisfying assignment yields a concrete mapping and its event pattern. A constraint then excludes that entire event-equivalence class, and the process repeats. This projects directly onto event classes rather than explicitly enumerating full mappings or permutation groups.

For a window $E \leq K$, decoding is complete when every retained family is certified exhausted or excluded. A timeout leaves completeness unresolved. This certificate concerns the saved search output within the window; it does not establish exhaustive exploration of all chemically possible mappings. The default deduplication preserves families that contain different event counts, as illustrated in the heavy-atom projection in Figure 1c.

3 Evaluation

3.1 Datasets and configurations

We evaluate annotated mapping recovery on all 1,851 Golden records (8) and event alternatives on 140 native coordinate/WBO endpoint pairs. Golden inputs use formal bond orders and explicit-H expansion. Reference labels are removed before search. Correctness compares the annotated heavy-atom correspondence, including unmatched atoms, modulo endpoint chemical automorphisms preserving charge, isotope, hydrogen count, and stereochemistry. A recovery requires a verified reference witness among the returned alternatives. Unresolved cases remain in the denominator.

We use two explicit prior-method baselines. LocalMapper is a prior accuracy state-of-the-art method as reported by Chen et al. (2); it is also the strongest single-output mapper in our strict Golden re-evaluation (1,605/1,851, 86.71%). Its published Golden accuracy was 89.8% on 1,758 retained reactions with manual calibration, so we keep the literature result separate from our rescored result. SLAP with our bidirectional edge-sweep wrapper is the strongest evaluated prior-method configuration for recovery among returned alternatives (1,796/1,851, 97.03%). It is the paired baseline for directional recovery and same-host search timing. The sweep wrapper is our evaluation extension, not a published SLAP state-of-the-art score. Runtime baselines are identified within each recorded hardware and output scope rather than assigning one method a universal speed ranking.

The coordinate collection contains balanced endpoint pairs with 18–137 explicit atoms, including organometallic and multicomponent structures. It has no annotated mappings and was inspected during method development. This is a bond-event comparison, not a mapping-accuracy benchmark: the mappings are unverified. We compare the minimum signed-event count reached by each method and the alternative event patterns at that count. These are minima within the returned candidate sets, not proven global optima or validated mechanisms. SLAP outputs are rescored on the same original WBO matrices using Equation (5). Each

Table 1 Evaluation configurations. Both use $b = 0.2$, $\tau_{\text{iso}} = 1$, explicit H, and uncut plus single-edge searches. The default is one seed per cut with the sweep; no-sweep controls are ablations retaining only the uncut graph. A seed setting counts source-atom orderings per cut.

Setting	Golden	Coordinate/WBO
Reactions	1,851	140
Search directions	Both, unioned	Reactant to product
Seed orderings	1 (default), 2, 3, 10	1 (default)
Branch cap	100	2,000
Competition completions per reaction	0	Up to 128
Evaluation	Reference-family recovery	Signed event-class coverage

returned heavy-atom assignment is completed by symbolic hydrogen refinement. The comparator classes are those observed among these returned witnesses; its hydrogen label families are not exhaustively decoded.

[Table 1](#) separates the settings actually evaluated. Golden measures the fragment-search stage; the 140-case evaluation also includes local competition and final event decoding. The competition configuration considers up to eight distinct parent partitions and two successive priority changes. Its budget was selected on development cases before the full 140-case evaluation. Comparator mappings are not used to propose or accept search branches. A two-order run of the final coordinate pipeline was not performed. Detailed protocols and artifact identifiers are given in [Appendix A](#).

3.2 Golden mapping recovery

With the default one ordering per cut and single-edge sweep, bidirectional GRAFT recovers 1,834/1,851 annotated mappings (99.08%). Three orderings recover 1,837 (99.24%; [Table 2](#)), and ten recover 1,840 (99.41%), with all reference-recovery verdicts resolved. Two orderings also recover 1,834 (99.08%), with all reference-recovery verdicts resolved at these settings as well.

For the default setting, smaller-first search recovers 1,803/1,851 references (97.41%), larger-first search recovers 1,779 (96.11%), and their union recovers 1,834 (99.08%; [Table 4](#)). Size counts all explicit atoms, including H. On equal-size endpoints, smaller-first retains the original input orientation and larger-first reverses it. The union adds 31 recoveries over smaller-first alone. Among the 1,084 strictly unequal-size pairs, the corresponding counts are 1,055, 1,043, and 1,072; among 767 equal-size pairs, they are 748, 736, and 762. Thus the bidirectional gain includes both size regimes.

These are family-recovery measurements: they quantify whether a reference correspondence is retained, rather than whether it is ranked first. Verification processes saved cuts independently, retaining one cut in memory at a time. A reference is absent from the saved union only when every scheduled cut has been checked and excludes it; missing cuts or inconclusive checks remain unresolved.

In the no-sweep ablation, one-seed GRAFT recovers 1,489/1,851 references (80.44%), with no unresolved cases. Uncut SLAP, unioned over both directions and binary/weighted modes, recovers 1,661 (89.74%), with one unresolved case. Adding the sweep recovers 345 additional references for GRAFT and 135 for SLAP. Thus, GRAFT’s higher recovery in the swept comparison depends on the broader search: its one-seed uncut control has lower coverage than the uncut SLAP union. The controls use saved uncut outputs from the same final campaign; no new searches are needed.

The gain is consistent with GRAFT’s greedy fragment growth. Once an extension is accepted, placements incompatible with that larger fragment are discarded; branching over the surviving placements cannot recover them. For example, growing fragment A across an edge (u, v) can exclude a placement that would permit a different neighboring fragment B. Omitting (u, v) in a sweep run removes that adjacency from both growth and fragment-preservation tests, potentially changing where growth stops and which placements survive. The union of these runs therefore explores alternative fragment boundaries while retaining the uncut result. Each run remains greedy, and a single-edge sweep cannot explore every possible partition. Cut-specific seed

Table 2 Golden recovery with and without the sweep. Recovery uses all 1,851 records. CPU averages use the same 1,821 completed reactions on the Mac, including both directions and the respective sweeps; reference verification is excluded. Dashes mark unmeasured GRAFT ablation search cost or mixed-host configurations whose times are not compared. The SLAP no-sweep time uses only the saved uncut calls. SLAP unions binary and weighted modes in both directions; GRAFT uses both directions. "Absent" means excluded from saved families; "unknown" denotes unresolved verification.

Configuration	Recovered	%	Absent	Unknown	CPU s/reaction
GRAFT, 1 seed + sweep (default)	1,834	99.08	17	0	2.52
GRAFT, 2 seeds + sweep	1,834	99.08	17	0	4.95
GRAFT, 3 seeds + sweep	1,837	99.24	14	0	–
GRAFT, 10 seeds + sweep	1,840	99.41	11	0	–
SLAP sweep (prior baseline)	1,796	97.03	52	3	4.82
GRAFT, no sweep (ablation)	1,489	80.44	362	0	–
SLAP, no sweep	1,661	89.74	189	1	0.067

orderings also contribute trajectory diversity, so the measured gain reflects the complete sweep schedule. Final bond events are always evaluated on the original, uncut endpoint matrices.

The SLAP single-edge sweep recovers 1,796/1,851 references (97.03%) using both directions and binary and weighted modes. Relative to this strongest evaluated prior-method baseline, one-order GRAFT has a higher verified recovery rate by 2.05 percentage points. The paired comparison contains 43 references verified only by one-order GRAFT and 5 verified only by SLAP. This compares recovery among alternatives under the stated configurations; it is not a comparison of single-answer accuracy or equal-size candidate sets.

On the same-Mac subset of 1,821 completed reactions, default GRAFT uses a mean 1.14 CPU seconds for smaller-first search, 1.38 for larger-first, and 2.52 for the bidirectional union, compared with 3.33, 1.49, and 4.82 for the corresponding SLAP sweeps (Table 4 and figure 2). Two-seed GRAFT costs 2.23, 2.72, and 4.95 CPU seconds. Thus, orientation and seed count both affect the recovery–cost tradeoff. GRAFT’s default bidirectional median is 0.72 seconds versus SLAP’s 0.88, but its 95th percentile is higher: 12.91 versus 11.05 seconds. The default bidirectional mean is 0.52 times the SLAP-sweep baseline (47.7% lower), whereas two-seed GRAFT costs 1.03 times that baseline. Lower means and medians do not imply uniformly lower runtime.

The comparison excludes reference verification. GRAFT includes checkpoint I/O and symmetry finalization; SLAP includes graph construction, mapping, and export but excludes I/O. Bidirectional CPU is the sum of the two direction costs. SLAP’s paired no-sweep union costs 0.067 CPU seconds on the same cohort; GRAFT’s no-sweep search cost was not separately recorded. Three- and ten-seed searches span Mac and Linux hosts, so their times are not pooled into this comparison. Archived default-mapper call times, including RXNMapper, LocalMapper, and RDT, are reported separately in Table A3, with their cluster hardware and narrower call scope stated.

3.3 Comparison with released mapping implementations

We also evaluate default released implementations of RXNMapper (11), LocalMapper (2), Chython (3), Indigo (6), and the Reaction Decoder Tool (RDT) (10), together with default binary and weighted SLAP. Each receives all 1,851 label-free Golden records. Saved predictions are rescored with the same strict heavy-atom relation criterion, retaining unmatched atoms and checking endpoint chemistry. No reference-guided reranking, balancing, or prediction repair is applied.

Table 3 compares GRAFT with every benchmarked implementation using reference recovery over the evaluated output. GRAFT ranges from 1,489 recoveries without the sweep to 1,834 with a one-seed sweep and 1,840 with ten seeds. LocalMapper recovers the strict reference relation in 1,605 reactions, RXNMapper in 1,547, and Chython in 1,561. Default binary SLAP recovers 1,591 among its explicit candidates, compared

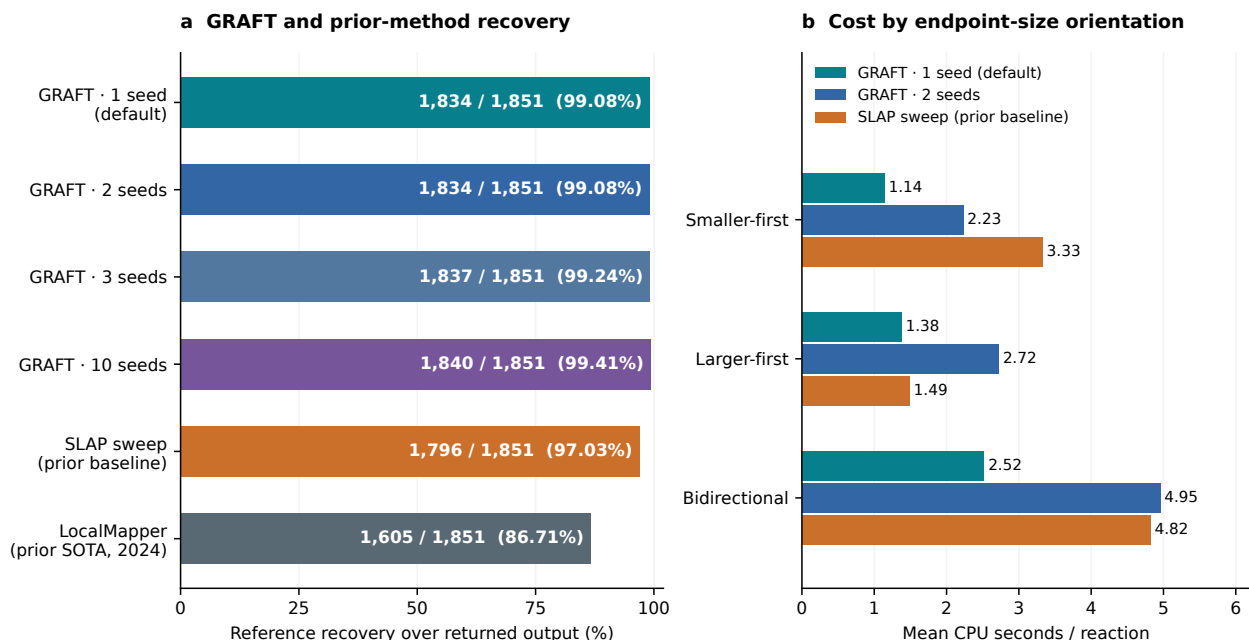


Figure 2 Reference recovery and search cost. (a) Golden recovery for swept GRAFT and SLAP, with LocalMapper as the prior accuracy-SOTA baseline (2024). LocalMapper returns one mapping; the other bars measure coverage among alternatives. All values use our evaluator, not literature-reported accuracies. Unswept controls are reported in Table 2. Every percentage uses all 1,851 records. (b) Mean CPU for smaller-first, larger-first, and bidirectional search on the same 1,821 completed reactions on the Mac. Reference verification is excluded; the procedures have different search objectives, output cardinalities, and I/O timing scopes.

with 1,796 for the broader bidirectional sweep evaluated above. These default calls use different search budgets from GRAFT; GRAFT’s family recovery is not a top-one accuracy measurement.

RDT recovers 1,030 references under this relation-based criterion, which differs from its chemical-change scoring. Indigo’s 692 recoveries require particular care: 503 outputs contain duplicate heavy-atom map labels, seven alter endpoint chemistry, and 133 calls fail. Chython changes endpoint chemistry in 29 outputs, and three RDT outputs fail validation. These results characterize the pinned implementations and input/output protocol rather than establish universal rankings of the methods. Versions and reproduction details are given in Appendix A.2.

3.4 Minimum bond-event comparison on 140 coordinate reactions

Default one-seed GRAFT with fragment competition reaches a lower minimum signed-event count than the SLAP sweep in 4 reactions, the same count in 136, and a higher count in none (Table 5 and figure 3). Against native SLAP, GRAFT reaches fewer events in 15 reactions and ties in 125. The four differences against the sweep are cases 35, 52, 69, and 94: each differs by one event. Because the mappings are unverified, fewer events indicate a difference between returned correspondences, not greater chemical accuracy.

Matching the minimum count does not imply recovering every pattern at that count. On the 136 reactions where the GRAFT and SLAP-sweep minima agree, GRAFT retains 162 of 164 comparator minimum-event patterns, covering every compared pattern in 135 reactions. Case 64 contains the two missing alternatives: GRAFT retains three of the five four-event patterns returned by the sweep. Against native SLAP, all 140 minimum-event patterns are retained on the 125 reactions with equal minima. GRAFT returns 171 distinct patterns at its own per-reaction minima, compared with 160 for native SLAP and 168 for the sweep; these counts describe alternative outputs, not verified mappings.

The larger decoding windows answer a different question: whether GRAFT retains a comparator pattern even when GRAFT also finds fewer-event alternatives. Each window ends one event above the larger native-SLAP

Table 3 GRAFT and benchmarked mapping implementations on Golden. Reference recovery uses all 1,851 reactions, including failed and unresolved cases. GRAFT is evaluated over its compressed families; SLAP over its returned explicit candidates, without a ranking or exhaustive decoding of its label families; the remaining methods return single mappings. GRAFT uses both directions; SLAP “union” combines both directions and binary/weighted modes. Search budgets and output sizes differ, so this is coverage of returned alternatives, not top-one accuracy. †: LocalMapper, prior accuracy SOTA reported in 2024; ‡: the SLAP sweep, strongest evaluated prior-method alternative-coverage baseline. Versions are specified in [Appendix A.2](#).

Method	Search setting	Evaluated output	Reference recovered
GRAFT	1 seed, sweep (default)	Families	1,834 (99.08%)
GRAFT	2 seeds, sweep	Families	1,834 (99.08%)
GRAFT	3 seeds, sweep	Families	1,837 (99.24%)
GRAFT	10 seeds, sweep	Families	1,840 (99.41%)
GRAFT	1 seed, no sweep (ablation)	Families	1,489 (80.44%)
SLAP, binary	Default, no sweep	Candidates	1,591 (85.95%)
SLAP, weighted	Default, no sweep	Candidates	1,564 (84.49%)
SLAP, union	No sweep	Candidates	1,661 (89.74%)
SLAP, union [‡]	Sweep	Candidates	1,796 (97.03%)
RXNMapper	Default, no sweep	Mapping	1,547 (83.58%)
LocalMapper [†]	Default, no sweep	Mapping	1,605 (86.71%)
Chython	Default, no sweep	Mapping	1,561 (84.33%)
Indigo	Default, no sweep	Mapping	692 (37.39%)
RDT	Default, no sweep	Mapping	1,030 (55.65%)

Table 4 Recovery and runtime by endpoint-size orientation. All rows use the sweep. Recovery uses all 1,851 Golden records; mean, median, and 95th-percentile CPU seconds per reaction use the same 1,821 completed Mac reactions. SLAP is the prior-method baseline and unions binary and weighted modes within each direction. CPU/SLAP divides each mean by the SLAP mean for that same orientation; values below one indicate lower mean cost. Size counts explicit atoms; equal-size ties use stored order for smaller-first and its reverse for larger-first. Bidirectional cost sums both searches, not parallel wall latency. GRAFT includes checkpoint I/O and symmetry finalization; SLAP includes graph construction, mapping, and export but excludes I/O. Reference verification is excluded.

Method	Orientation	Recovered	Mean	Median	95th pct.	CPU/SLAP
GRAFT, 1 seed (default)	Smaller-first	1,803 (97.41%)	1.141	0.263	5.289	0.34
	Larger-first	1,779 (96.11%)	1.377	0.348	6.401	0.92
	Bidirectional	1,834 (99.08%)	2.518	0.719	12.913	0.52
GRAFT, 2 seeds	Smaller-first	1,815 (98.06%)	2.232	0.416	10.635	0.67
	Larger-first	1,805 (97.51%)	2.721	0.578	14.064	1.83
	Bidirectional	1,834 (99.08%)	4.953	1.250	26.350	1.03
SLAP sweep (prior baseline)	Smaller-first	1,775 (95.89%)	3.329	0.422	4.752	1.00
	Larger-first	1,770 (95.62%)	1.490	0.455	4.568	1.00
	Bidirectional	1,796 (97.03%)	4.819	0.881	11.048	1.00

and SLAP-sweep minimum. Within these windows, GRAFT retains 155/160 native-SLAP and 166/168 SLAP-sweep minimum-event patterns. The 336 total decoded classes include higher-event alternatives; only 171 lie at GRAFT’s own minima. Competition adds 36 classes in 9 reactions within these windows, including one class requiring internal family shuffles. Complete window decoding concerns the saved families and does not certify exhaustive structural search.

Table 5 Minimum bond-event comparison on 140 coordinate reactions. Counts compare GRAFT’s per-reaction minimum with the stated comparator minimum. The last column compares patterns only where the two minimum counts agree. All mappings are unverified; this is not a mapping-accuracy benchmark. GRAFT uses its default one-seed sweep with competition. Native SLAP is the released prior-method baseline; SLAP sweep is its extended search control.

Compared with	GRAFT fewer	Equal	GRAFT more	Patterns at equal minima
SLAP sweep	4	136	0	162/164
Native SLAP	15	125	0	140/140

Table 6 Recorded stage costs for the 140 coordinate reactions. CPU seconds per reaction on the Mac. GRAFT uses one seed, cap 2,000, and the sweep. Its decoder exhausts the full event windows, whereas SLAP supplies returned witnesses; these are different outputs, so the table does not give a minimum-only or end-to-end speed ranking. Search/-competition values use the successful recorded stages; decoding includes all continuation passes. Final shared SLAP event rescoring and failed or replaced attempts are not timed here. SLAP is the prior-method mapping baseline; no comparable prior full-family decoder timing is available.

Recorded stage	Mean CPU s	Median CPU s	95th pct. CPU s
GRAFT sweep search	1.047	0.314	3.481
GRAFT fragment competition	1.840	1.007	4.825
GRAFT full-window decoding	11.014	1.224	52.668
SLAP sweep mapping	1.337	0.546	5.919
SLAP H refinement / initial scoring	0.053	0.011	0.207

GRAFT’s ordinary sweep averages 1.05 CPU seconds per reaction, competition 1.84, and full-window decoding 11.01; their medians are 0.31, 1.01, and 1.22 seconds (Table 6). SLAP’s binary bidirectional sweep averages 1.34 CPU seconds for mapping and 0.053 for hydrogen refinement and initial scoring. The SLAP mapping timer includes cut construction and mapping calls but excludes input preparation and export; the refinement timer includes its recorded I/O. Both methods’ event patterns are compared on the original WBO matrices with the same signed-event thresholds. Native-only SLAP time and a GRAFT minimum-only decoding time were not separately recorded.

3.5 Final representation and postprocessing cost

Unordered fragment-pair deduplication reduces the 140-case total from 237,645 ordered branch records to 124,641 final branches, a 47.55% reduction. The median falls from 596.5 to 397 branches per reaction. The largest individual reduction is from 52,669 to 2,856.

The final representation retains 236,653 distinct normalized mapping-family records. Thus, fewer branch identities do not imply an equal reduction in event-decoding work: several constrained families can belong to the same fragment combination. Catalogue construction, shared intrinsic symmetry computation, representative checks, and complete event-window decoding together take 11.60 minutes of wall time and 25.70 recorded CPU-minutes on three local workers. Peak combined sampled memory is 1.88 GiB. These measurements include certificate journaling and bounded continuation, and exclude matching and competition. Per reaction, the mean cost is 11.01 CPU seconds, the median is 1.22, and the 95th percentile is 52.67, summing all continuation passes for each case. Other benchmark jobs shared the computer, so the wall time is an observed campaign duration rather than an isolated latency measurement.

Every family in all 140 catalogues was freshly decoded or certified within its specified event window (Appendix A). The output contains 336 distinct reaction-specific event classes, each with a mapping witness. No case remains unresolved within those windows. Preservation of the family union follows from the transformation in Equations (3) and (4).

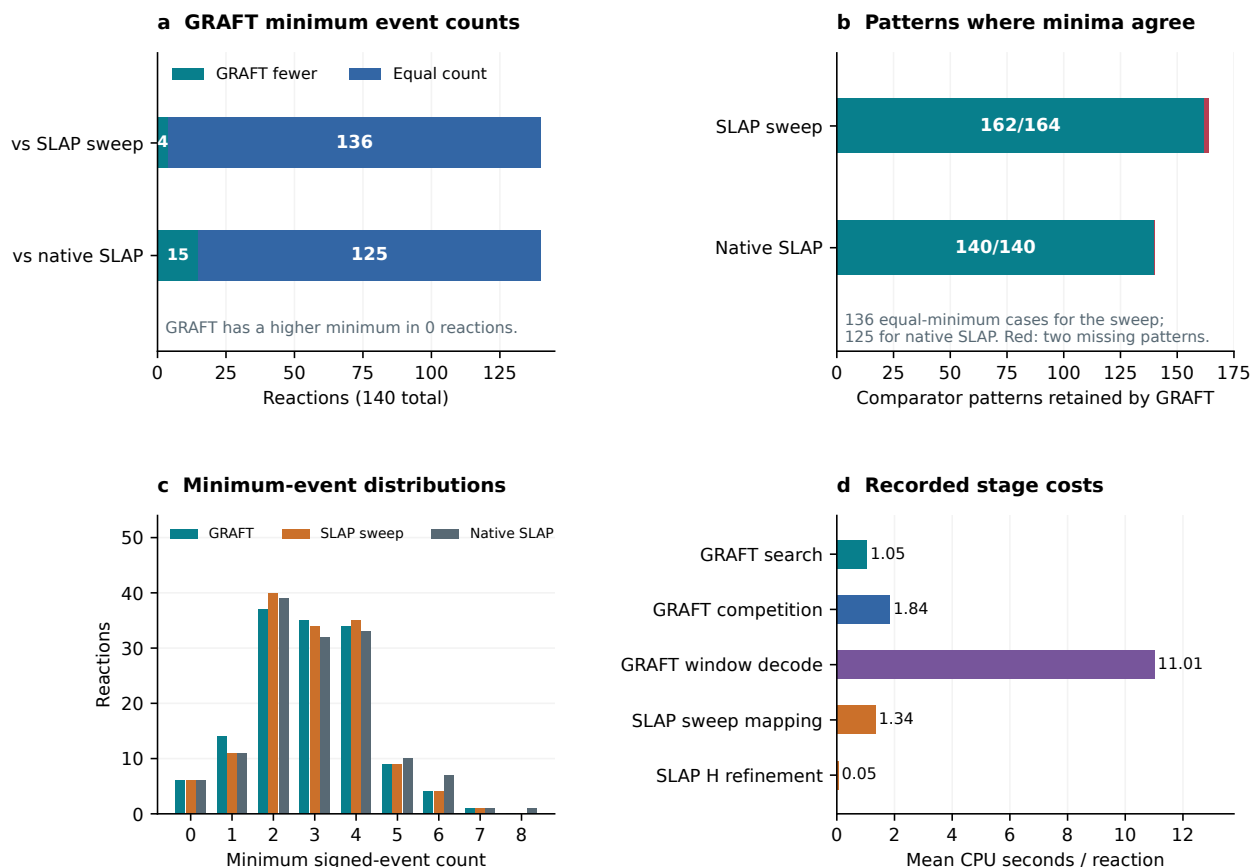


Figure 3 Minimum bond events and measured costs on 140 coordinate reactions. Mappings are unverified: this compares returned bond-event patterns, not mapping accuracy. GRAFT uses the default one-seed sweep with competition. (a) Reactions where GRAFT reaches fewer or equal events than the comparator; it reaches more in none. (b) GRAFT coverage of comparator minimum-event patterns, restricted to reactions with equal minima. The missing sweep patterns are both in case 64. (c) Per-reaction minimum-event distributions for GRAFT, native SLAP, and SLAP sweep. GRAFT has 171 minimum-event patterns, compared with 160 and 168 for the two prior-method configurations; higher-event alternatives are excluded. (d) Mean CPU per recorded stage on all 140 reactions. GRAFT decoding exhausts full windows; SLAP refinement covers its returned witnesses. Timing scopes differ as specified in Table 6.

4 Discussion

GRAFT separates structural search, symmetry representation, and interpretation of bond changes. Continuous growth accumulates context while preserving multiple placements; local competition allows neighboring fragments to challenge an established partition. Final grouping removes irrelevant growth order, while the flat family representation retains constraints and correlated permutations that can affect the output. Symbolic projection then exposes distinct bond-event hypotheses without expanding every mapping.

4.1 What a single correspondence leaves out

One correspondence is sufficient when atom identities are established and the downstream quantity is invariant over the remaining alternatives. Otherwise, selecting one witness discards two kinds of information: different assignments can produce different bond-event patterns, and a chosen pattern can still admit joint atom shuffles. Formally, if $\mathcal{F}$ is the retained mapping set and $e(m)$ its event-class projection, a witness m_0 supplies $e(m_0)$ but cannot reconstruct $\{e(m) : m \in \mathcal{F}\}$ whenever that set has more than one member. Event counts alone are also insufficient: distinct signed patterns can have equal counts. GRAFT therefore returns an event-class representative together with access to its supporting families and conditional shuffle queries.

Golden case 9 illustrates why retaining alternatives matters (Figure 4). GRAFT recovers the annotated heavy-atom correspondence: the alcohol oxygen O₁₅ enters the product ring, while the ketone oxygen O₃ leaves in water (A). A second recovered mapping exchanges these oxygen origins (B), and a third additionally exchanges the correspondence of C₁₃ and C₁₄ (C). The three displayed witnesses have five, six, and seven heavy-atom changes, respectively; including H gives nine, eight, and nine signed events. Each witness attains the minimum number of H-involving changes for its fixed heavy-atom correspondence, as verified from the attached-H counts. Thus a lower total event count can favor a nonreference mapping, while equal counts (A and C) do not establish equal correspondences. Golden annotates heavy atoms, not H identities or a validated pathway. These are selected recovered witnesses, not an exhaustive decoding of this reaction.

The output distinction is specific to the interfaces evaluated here. RXNMapper, LocalMapper, Indigo, Chython, and RDT returned at most one explicit candidate record per reaction in the archived runs; SLAP exposed multiple records and compressed labeled-graph information, while GRAFT exposes conditional mapping families. This does not establish that only SLAP and GRAFT can return alternatives. For example, Ali et al. enumerate optimal mappings with an Ising-computing approach and reduce them by molecular symmetry (1). Nor does high reference recovery establish near-complete coverage of all mappings: GRAFT’s structural search is bounded, and complete decoding certifies only the saved families within the requested event window.

4.2 Implications for pathway and transition-state analysis

Different bond-event patterns can suggest different reaction centers and alternative transition-state starting guesses. They do not identify pathways uniquely: a single net atom mapping can be compatible with several intermediate sequences, and an alternative mapping can be chemically implausible. Energies, reaction conditions, stereochemistry, and isotope tracing can distinguish possibilities that endpoint connectivity alone cannot. Prior work on reaction-pathway search also identifies atomic-index assignment as a practical obstacle and addresses the factorial number of index assignments without enumerating them (4).

For a transition-state pipeline, the relevant choices are the event-class candidate and the allowed atom assignments around its reaction core. Define the core as the atoms incident to the candidate’s signed bond events. Alternative assignments of those atoms can be queried jointly against the saved family constraints, retaining only assignments that extend to a complete mapping and satisfy the chosen event-class restriction. Required correlated movements of other atom assignments must be included in that completion. The completed mapping determines the product atom ordering supplied to the interpolation step.

Different completed mappings can then give different interpolations between the same endpoint structures. For illustration, in a common coordinate frame, a Cartesian interpolation for mapping m is

$$\mathbf{x}_i^{(m)}(t) = (1 - t)\mathbf{x}_i^R + t\mathbf{x}_{m(i)}^P, \quad 0 \leq t \leq 1. \quad (6)$$

Changing $m(i)$ changes the product coordinate assigned to reactant atom i , and hence generally changes its interpolated trajectory. Retaining multiple event classes and their allowed core assignments therefore lets a downstream pipeline construct and assess several initial paths instead of committing to one correspondence before interpolation. Equivalent relabelings applied consistently to an entire calculation do not create new physical pathways; the issue is the relative correspondence between fixed endpoints. These examples motivate retaining alternatives and accessible symmetry information; transition-state search performance is not evaluated here.

Golden case 9: the reference is one of several recovered mappings

GRAFT · 1 seed + cut sweep (default) · branch cap 2,000 · iso tolerance 1 · event threshold 0.5

● ● ● Saved fragment groups

⊗ Break / weaken

⊕ Form / strengthen

A Ground truth (Golden reference) — recovered by GRAFT

5 heavy-atom changes



B GRAFT alternative — different oxygen retained

6 heavy-atom changes



C GRAFT alternative — oxygen and carbon correspondence change

7 heavy-atom changes



Retaining alternatives matters: a lower total event count (B: 8) need not select the annotated mapping (A: 9).

Ground truth refers to Golden's heavy-atom annotation. H identities are unannotated; totals use each GRAFT witness.

Numbers are original Golden labels. H atoms are implicit. Selected witnesses do not imply exhaustive or minimum-event decoding.

Figure 4 Golden reference and recovered GRAFT alternatives for case 9. A matches all 15 annotated heavy-atom pairs exactly. B exchanges the oxygen origins; C also exchanges C₁₃/C₁₄. GRAFT uses one seed with cut sweep (default), a 2,000-branch cap, and $\tau_{\text{iso}} = 1$; reference labels are removed before search. Pastel colors identify saved fragment groups, consistently across each row's endpoints; partitions can differ between rows. Solid black lines retain the endpoint bonds. Separate $\otimes$ (break/weaken) and $\oplus$ (form/strengthen) badges point to changed bonds without covering the fragment colors. Events use formal bond orders at threshold 0.5. Numbers are the original Golden atom labels. H atoms are implicit; the figure separates heavy-atom changes from totals including H. "Ground truth" denotes the dataset's heavy-atom annotation, not a verified pathway. Only three selected witnesses are shown.

4.3 Coverage and remaining search limitations

The Golden results support broad reference coverage by the fragment-search stage, and the coordinate comparison shows that GRAFT reaches equal or lower event counts than the returned SLAP-sweep candidates while retaining most patterns at shared minima. These are complementary measurements. The coordinate collection has no mapping annotations, and event minimization is not an energy model or a proof of mechanistic plausibility. The priority budget was tuned on development cases, so independent validation is needed to establish generalization.

An audit of all eleven ten-order Golden misses separates unmatched-atom decisions from fragment-placement failures (Table A2). In four reactions, every retained family maps exactly one more heavy-atom pair than the reference. Such a family cannot reproduce the reference's unmatched-atom pattern through symmetry shuffling: permutations preserve the number of assigned pairs. The remaining seven lack the required correspondence within their saved families. Re-ranking or expanding those families into explicit bijections cannot supply a missing structural alternative.

Two traced examples show how greedy extension excludes a representable reference. One reference exchanges adjacent labels in an otherwise unchanged pyridine ring; another exchanges the carbonyl-oxygen attachments of oxalic acid across its retained carbons. Ordinary growth preserves bonds that these correspondences require separating. Uncapped trajectories still miss both references, while reference-constrained fragment choices establish their representability. In the oxalic-acid example, a reference-selected three-edge relaxation also exposes the reference to ordinary matching; the single-edge sweep does not contain that cut set. These are feasibility diagnostics, not additional benchmark recoveries. For the other five correspondence misses, the audit does not isolate cap truncation from unsampled growth choices. They motivate selective alternatives to successful growth extensions and explicit choices about leaving atoms unmatched, rather than treating larger caps or more symmetry expansion as sufficient remedies.

Only one of the eleven misses is recovered by any tested comparator configuration, including the SLAP sweep. The shared failures warrant inspection of the supplied correspondences as well as search improvements; they do not establish annotation errors. All eleven remain in the reported denominator. A complete event-window certificate likewise applies only to the saved families, whose matching-preserving automorphisms can still change finer-threshold events.

Code and data

Source code, benchmark summaries, and figure-generation scripts accompany the repository at https://github.com/yunhzou/coordinate_alignment, branch `paper/continuous-fragment-growth`. Golden is distributed by its original authors (8). Per-case results and the configurations used for each measurement are supplied with this manuscript.

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A Reproducibility details

A.1 Search and comparison protocols

The active-edge floor is $b = 0.2$ and the matching tolerance is $\tau_{\text{iso}} = 1.0$. The default search uses one seed ordering per cut with the sweep; additional seeds and the no-sweep ablation are explicitly labeled. Seed streams are deterministic with root seed 42. Sweeps retain the uncut search and individual source-edge deletions. Bidirectional Golden output is a union of returned relations; branches from opposite directions are not spliced. Growth and branch caps bound compressed candidates rather than explicit mapping counts.

Golden uses dataset revision 793475e. Its heavy-atom evaluator compares correspondence relations modulo endpoint chemical symmetry, retains unmatched atoms, and rejects altered endpoint chemistry. Hydrogen atoms participate in search but their individual identities are not reference labels. All 1,851 records remain in the denominator. All reported GRAFT configurations were rerun against the final algorithm source, commit 3a9ef9a. Execution manifests and component hashes accompany the per-case results. Golden evaluates the fragment-search stage, while the coordinate protocol also evaluates competition and event decoding.

The no-sweep GRAFT control retains only the original uncut checkpoint at one seed in each direction. Its 3,702 saved graphs are finalized and evaluated using the same reference-family criterion, yielding 1,489 recoveries and 362 completed non-recoveries. For SLAP, the control retains ordinal-zero calls from both directions and both bond modes; incomplete sweep records are checked at that ordinal separately. Case 1723 has no committed reverse uncut output and remains unresolved. These are paired controls of the final campaign, not substituted historical configurations. Re-evaluation time is excluded from search-cost claims.

Each search or evaluation subprocess has a 300-second watchdog and an initial 3 GiB sampled-memory limit. Externally interrupted phases can resume their completed cuts for at most two bounded continuations with a 6 GiB allowance. Golden reference evaluation has a 220-second internal budget. Completed raw cuts can also be finalized and verified individually, without new search. A positive result requires a verified witness; it can establish recovery even when another direction or the combined archive is incomplete. A negative result requires completed verification of the saved output; other cases remain unknown.

The campaign begins on an Apple M4 Pro Mac with 14 logical CPUs and 48 GiB RAM, using at most eight single-core workers, and continues on Linux CPU nodes with a final concurrency limit of 150 single-thread workers. Python 3.12.14, RDKit 2026.3.6, pynauty 2.8.8.1, SymPy 1.14.0, and Z3 5.1.0.0 are fixed across hosts. Native extensions are rebuilt on Linux; source-hash checks, focused tests, and two search/reference parity cases verify the port. Numerical-library threads are fixed to one. CPU averages use 1,821 common reactions with both one- and two-order searches completed on the Mac and complete comparator output. Search-call timing includes checkpoint I/O and excludes imports, preparation, reference evaluation, and interrupted work; SLAP’s instrumented workflow excludes I/O. Higher-order searches use mixed hosts and are excluded from this timing comparison.

The SLAP comparator uses SLAPMapper 1.0.0, revision ea248fd, with both directions and binary and weighted modes. Each sweep variant uses one original ordering and rebuilds initialization for the uncut input and each single-edge deletion. Returned labels are restored to the original endpoint structures before evaluation. Its 1,796 verified Golden recoveries leave 52 completed misses and 3 unresolved records; all remain in the denominator. The sweep perturbs SLAP’s input graph and assignment objective, whereas GRAFT relaxes fragment-preservation constraints.

On the coordinate collection, the exact original XYZ-derived SLAP adjacency matrices are restored from the saved graph records and verified against every source-edge deletion. Atom order and atomic numbers are retained; mapping labels are reset. Fresh native and bidirectional single-edge SLAP runs use binary graphs and the original initialization. The XYZ-to-graph preparation itself is not rerun. GRAFT uses the cached WBO matrices as supplied. The available benchmark metadata do not specify their electronic-structure method or charge/spin preparation; these matrices therefore define the computational inputs, while regeneration from molecular structures is not established by this evaluation. Each returned SLAP heavy-atom assignment receives a symbolic hydrogen refinement under the comparator’s original bond-change objective, and its witness is rescored on the original WBO matrices using Equation (5). Tied hydrogen optima can give different signed-event classes; the reported comparator set comprises the returned witnesses, not the

entire hydrogen label relation. This compares the stated endpoint workflows rather than isolated solvers on identical graph representations.

A.2 Default comparator reproduction

The default comparator predictions were generated with RXNMapper 0.4.3, LocalMapper 0.1.5, Chython 2.18 with chython-rxnmap 2.0, Indigo 1.46.0, SLAPMapper 1.0.0 at revision [ea248fd](#), and RDT 4.0.0 at revision [0d663225](#). Chython’s released ONNX model is identified by its saved hash; it is not represented as a reproduction of a historical GraphormerMapper checkpoint. RXNMapper and LocalMapper use their released default models on CPU. SLAP is evaluated over all returned explicit candidates, without ranking, with its default heavy-atom symmetry handling, with binary and weighted settings reported separately. Indigo uses its native automapping interface; RDT uses the public `RDT.map` API.

The complete archive retains mapping strings, explicit candidate order, version/model hashes, and per-record evaluations. All 12,957 method/reaction records were rescored with the current evaluator and agree with the archived outcomes. Dataset and label-free input hashes match the final GRAFT campaign. Agent-field components returned by a mapper are retained on the input side, preserving their original atom labels. Changed endpoint chemistry and duplicate heavy-atom labels invalidate a prediction. All 1,851 records remain in each denominator.

RDT’s final records include execution retries after a worker failure; every final call returns an output. Invalid outputs are not repaired. These default implementations were timed on the original cluster, so their archived call times are not included in the same-Mac comparison in [Figure 2](#). [Table A3](#) reports their archived per-call timings separately; full timing and failure records accompany the reproducibility archive.

A.3 Bounded fragment competition

The coordinate configuration uses one seed ordering, cap 2,000, up to eight lowest-event distinct parent source partitions, at most two successive priority changes, and at most 128 proposal-growth calls and 128 completion calls per reaction. Smaller challenger fragments are visited first, with deterministic contact order. A proposed challenger is rematched with a single contacting source atom. Conflicting source assignments and occupied target images are released; attached components of at most eight atoms can also be released around the affected region. Remaining assignments are anchors for completion.

The admission check evaluates every complete representative against the original endpoint weights and its required fragment constraints. A completion graph is excluded if any checked complete representative violates these constraints. Only accepted graphs contribute to reported coverage. The ordinary search output is retained in union. Pending queue entries, omitted parent partitions, and capped continuations are not covered by decoding certificates.

Competition and decoding use five-minute per-case watchdogs with persisted results. Complete family certificates survive continuation, and unfinished families remain explicit. The full coordinate search, competition, and decoder stages are rerun from the original endpoint inputs. Timing for complete final decoding includes all of its bounded passes; it excludes the preceding search and competition stages.

A.4 Event windows and output equivalence

A signed event class contains the nonzero atom-pair signs from [Equation \(5\)](#), modulo source permutations preserving the full event-response coloring relative to the opposite endpoint. This coloring retains every threshold comparison needed to transport a pattern without changing its identity. It is separate from the intrinsic fragment matching symmetry in [Equation \(2\)](#). The exported witness records both the atom mapping and the formed/strengthened and broken/weakened pair lists.

The 140-case comparison uses fixed per-case windows from the evaluation protocol. Each limit is one event above the larger of the native-SLAP and SLAP-sweep minima under the common signed-event definition. All 236,653 retained flat families are exhausted in these windows without a class-count cap; higher-event patterns are outside the completeness claim.

Table A1 Complete final decoding on all 140 reactions. Every retained family is certified within the listed per-case event limit.

Maximum events	1	2	3	4	5	6	7	8	9
Reactions (all complete)	6	11	39	32	33	10	7	1	1

For the saponification example in Figure 1, the direct assignment $(1, 2, 3, 4, 5, 6) \mapsto (a, b, c, d, e, f)$ loses O4–C5 and gains C5–O6. Exchanging the images of O3 and O4 additionally weakens C2–O3 and strengthens C2–O4: two versus four heavy-atom events. Both assignments preserve the internal matching constraints of fragment $\{1, 2, 3, 4\}$ at $\tau_{\text{iso}} = 1$. The challenger instead assigns O4 to f and O6 to d, retaining O4–C5 and replacing C2–O4 by C2–O6. The original assignment transfers hydroxide oxygen into methanol; the challenger retains the methoxy O–C bond. Formal bond orders define this illustration; the displayed event counts include only heavy atoms. Exchanging the acetate oxygens also changes the localized resonance representation, so the event signatures do not establish distinct mechanisms. Both product acetate oxygens have zero attached hydrogens. The figure builder verifies these mappings and event lists. For the separate alcohol-site tree, it verifies every attached hydrogen and every bond inside each local matched fragment, the distinct target-site environments, and injectivity along all six displayed paths.

A.5 Lossless flattening

For each complete search path, flattening retains its terminal representative, the union of required fragment-bond constraints, and the allowed action sequence with prefix locks already applied. It removes only representation choices that leave the admissible relation unchanged. Reordering disjoint-support factors preserves their product; adjacent nested factors satisfy $HG = GH = G$ when $H \leq G$. Subgroup containment is checked by exact generator membership using a stabilizer chain. Other factors with overlapping support retain their order. Identical resulting records are shared, and final branch grouping stores the union of their family identifiers. Consequently, the union of mappings before and after grouping is the same. The procedure does not infer equality merely from a shared fragment partition or a shared automorphism group.

The event decoder obtains completeness by invariance certificates, valid lower bounds, or an unsatisfiability result after excluding every discovered output class. Budget exhaustion is recorded as incomplete. This distinguishes preservation of a saved mapping union, exhaustive decoding of a specified event window, and exhaustive atom-mapping search.

A.6 Audit of Golden nonrecoveries

We inspected every saved cut in both directions for the eleven ten-order misses, preserving the original reference and search outputs. Independent reconstruction from the original RDF agrees with the prepared reference in all eleven cases. Mapping cardinality is invariant within each saved family. For six of the seven remaining cases, necessary source/target orbit-pair counts exclude the reference even after pooling permitted target actions into an overapproximation; case 1285 requires the full joint-correspondence check. These tests concern the retained output, not every possible search trajectory.

Table A2 The eleven Golden nonrecoveries. Case indices are zero-based dataset indices. Retained pair counts cover every terminal in both directions and every saved cut at ten seed orderings and cap 100. The four cardinality mismatches concern exact reference relations, including unmatched atoms.

Case	Reference pairs	Retained pairs	Observed mismatch
7	29	30	One extra matched pair
19	23	23	Atom pairing excluded by orbit bounds
590	31	31	Atom pairing excluded by orbit bounds
871	20	20	Atom pairing excluded by orbit bounds
986	16	16	Atom pairing excluded by orbit bounds
1228	12	12	Atom pairing excluded by orbit bounds
1285	14	14	Joint C/O correspondence excluded
1358	36	37	One extra matched pair
1377	22	23	One extra matched pair
1475	18	19	One extra matched pair
1553	20	20	Atom pairing excluded by orbit bounds

All eleven cases have cap stops in at least one direction, which does not establish cap truncation as their cause. Separate ten-order sensitivity checks on cases 986 and 1285 at cap 1,000 or matching tolerance 0.5 remain nonrecoveries; several of those directional runs are uncapped. In the causal traces, ordinary Python and native graphs agree and are uncapped at cap 2,000. Reference-constrained matching represents all 16 and 14 annotated pairs, respectively. The successful case-1285 cut set omits two C–O attachments and one O–H bond; it is reference-selected, and neither its success nor the constrained witnesses contributes to the blind benchmark score. The audit, controls, traces, and source identities are archived with the benchmark report.

A.7 Default-comparator call times

Table A3 Archived default-mapper call times on the original Linux cluster. The calls column counts completed mapping calls, including outputs subsequently judged invalid; exceptions are excluded from these time averages. Recovery uses all 1,851 reactions and the same strict criterion as Table 3. † marks LocalMapper, the prior accuracy-SOTA baseline (2024). These are the released default settings without a sweep. Times exclude startup/model loading, queueing, reference evaluation, and saving. They are not comparable to the Mac sweep times as a hardware-controlled speed ranking.

Implementation	Recovery	Calls	Mean wall s	Median wall s	Mean CPU s
RXNMapper	83.58%	1851	0.056	0.052	0.056
LocalMapper [†]	86.71%	1851	0.294	0.265	0.294
Chython	84.33%	1846	0.080	0.073	0.612
SLAP binary	85.95%	1851	0.075	0.037	0.075
SLAP weighted	84.49%	1851	0.058	0.033	0.058
Indigo	37.39%	1718	0.184	0.008	0.184
Reaction Decoder Tool (RDT)	55.65%	1851	0.295	0.141	1.021

LocalMapper, the prior accuracy-SOTA baseline, has 86.71% strict recovery here and averages 0.294 seconds per call in both wall and CPU time. RXNMapper has the lowest recorded mean call time (0.056 seconds), with 83.58% strict recovery; speed and recovery therefore select different baselines within these default calls. Mean wall time describes the observed duration of a mapping call; mean CPU time sums its recorded processor time. Multithreaded implementations can accumulate more CPU time than wall time. Search scope also differs: the learned mappers return one mapping, default SLAP returns explicit alternatives, and the GRAFT measurements in Table 4 include a complete sweep and compressed output construction. GRAFT’s unswept search cost was not separately instrumented, so rescoring its saved uncut outputs is not reported as search time. No mapping runs were repeated to construct these timing tables.

A.8 Output multiplicity and the discussion example

A read-only audit of all 1,851 archived records per default comparator found no multiple-record outputs from RXNMapper, LocalMapper, Indigo, Chython, or RDT. Binary SLAP returned multiple records in 522 reactions and weighted SLAP in 417; these are explicit record counts, not counts after expanding label symmetries or grouping events. Figure 4 uses the three saved minimum-event witnesses for coordinate case 127 and their supporting GRAFT families. No mapping search or event decoding was repeated. Each witness was rechecked against its saved family, and every event was recomputed from the original WBO matrices. The H_{10}/H_{11} exchange in panel (b) is feasible in the supporting family and preserves the exact event-class identifier. The figure retains all 23 atoms in these checks; spectator H atoms are implicit only in the drawing. For a conventional chemical depiction, bonds with WBO at least 0.5 are shown as single bonds, or double bonds when WBO is at least 1.4; this depiction does not determine event counts. The complete matrices, mappings, fragment partitions, and source hashes are in `coordinate_case127.json`. A separate six-atom example in `docs/AAM_SIMPLE.ipynb` illustrates the public API interactively.